

# TULSI GUPTA

AI/ML Engineer | Backend Developer

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## SUMMARY

Computer Science engineering student skilled in Python, Machine Learning, Generative AI, Django, with hands-on experience building AI-powered and backend applications. Strong foundation in DSA, OOP, REST APIs, databases, and deep learning, with practical experience using LangChain and LangGraph to develop LLM-based applications.

## TECHNICAL SKILLS

**Languages:** Python, C++, C, JavaScript, SQL, HTML, CSS

**Backend & APIs:** Django, FastAPI, REST APIs

**Databases:** MySQL, PostgreSQL, MongoDB

**AI / ML:** PyTorch, Scikit-learn, LangChain, LangGraph, RAG, LLM APIs, ChromaDB, Embeddings

**Tools:** Git, GitHub, Postman, VS Code, Jupyter Notebook, Docker

**Core CS:** Data Structures & Algorithms, OOPs, DBMS, Operating Systems, Computer Networks

**Automation:** n8n

## PROJECTS

### SkyRecall — AI-Powered Cloud Storage Platform & Near-Duplicate Image Detection

**Tech Stack:** React.js, Django REST Framework, PostgreSQL, Cloud Storage, Python, OpenCV, PyTorch | [Github](#)

- Built a full-stack AI-powered platform for cloud-based photo/video storage, organization, and intelligent media retrieval.
- Integrated computer vision models for object detection, face recognition, metadata extraction, and semantic image search.
- Implemented a hybrid pHash + deep-learning embedding approach for detecting near-duplicate images using Hamming and Cosine similarity.

### Space Biology Knowledge Engine — NASA Space Apps Challenge 2025

**Tech Stack:** React.js, Django, Python, Neo4j, LangChain, OpenAI API, NLP, Knowledge Graphs | [Github](#)

- Built an AI-powered research exploration platform over 608 NASA bioscience publications, enabling semantic search and automated research summarization
- Designed a Neo4j knowledge graph connecting publications, experiments, organisms, research topics, missions, and scientific findings for relationship-based knowledge discovery.
- Developed an interactive React + Django dashboard for exploring research trends, connections, and knowledge gaps across NASA studies.

### Smart QR Attendance System

**Tech Stack:** React.js, Node.js, Express.js, MySQL, REST APIs | [Github](#)

- Developed a full-stack QR-based attendance system using React.js and Node.js to automate student attendance and replace manual record-keeping. Implemented secure backend verification to validate each scan and store accurate attendance data.
- Implemented QR code generation and scanning workflows with server-side validation to verify attendance requests and prevent invalid or duplicate submissions.
- Designed and integrated RESTful APIs for attendance processing, student management, and communication between the React frontend and Node.js backend.

## EDUCATION

Malout Institute of Management and Information Technology

B.Tech, Computer Science & Engineering | CGPA: 8.13 / 10

Expected 2027

## HACKATHONS

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- **Smart India Hackathon (SIH)** — Collaborated in a team to develop technology-driven solutions for real-world problem statements under competitive hackathon constraints.
- **NASA Space Apps Challenge** — Developed an AI-powered space-biology research exploration platform involving semantic search, knowledge graphs, and scientific literature analysis.
- **Agri Innovation Program** — Collaborated on technology-based solutions addressing real-world agricultural challenges through innovation and problem-solving.

## ACHIEVEMENTS & CERTIFICATIONS

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- **5th Position among 220+ Teams** — Recognized for developing an efficient data structuring and visualization system, demonstrating problem-solving and data-engineering skills
- **NCC 'C' Certificate** — Demonstrated leadership, discipline, teamwork, and training excellence through structured training.

## INTEREST

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- Sketching and Creative Design • Travelling • Technology experimentation